M.A.R.V.E.L Design

CEGR 2154 Sophomore Design Laboratory

Submitted By: Ancient Grains Engineering
Date Performed: 10/11/2024
Date Due: 10/21/2024

Instructor: Shen-En Chen
Teaching Assistant: Soroush Roghani

Grade: _____

We, Ancient Grains Engineering, have committed no violations of the UNC Charlotte Code of Student Academic Integrity in preparing and submitting this report.

Signature: Noah Kent Date: 10/11/2024

Letter of Submission

Group 2: Ancient Grains Engineering

9201 University City Blvd.

October 21, 2024

To: Dr. Shen-En Chen and Soroush Roghani

Ancient Grains Engineering would like to submit a draft memo outlining three conceptual designs for the Modernization of Advanced Research on Vehicle Electrification and Locomotives Campus located in Gastonia, NC.

In this document, you will find three separate site plan concepts for the MARVEL Campus, each complete with its own overview and short analysis.

By comparing the strengths, weaknesses, opportunities, and risks of each development plan, a final design will be able to be selected in the near future. This will allow the project to move forward into detailed planning work for each individual aspect of the design.

If you have any questions or require any clarifications regarding our site concepts or analyses, please do not hesitate to contact our group.

Sincerely,

Noah Kent, CEO

Connor Schuh, CTO

Sam Dashti, CFO

Seth Hall, CCO

Gabriel Jugan, Engineer of Record

Executive Summary

The end product of this project is a UNC Charlotte satellite campus focused on the development and research of sustainable modes of transportation, particularly in the fields of electric vehicles and locomotives.

There will be three primary academic/research buildings, each with their own purpose. The Engineering Education Building (EEB) will serve as the primary location of lectures, offices, and study rooms. The Flexible Research Laboratory will serve as the primary location for studio-based research projects. The final building, RailSparc, will focus on railroad research, with its main attraction being three attached train depots with direct rain access.

Additionally, there will be a drone playground to allow for the testing of new drone technology. This will offer any students participating in drone research a convenient and regulated space to test their designs, as well as provide a recreational area for the surrounding community.

There will also be a water pond created with the primary purpose of conducting underwater research into underwater environments. Like the drone playground, this will be a close and controlled environment for students working in the field to get practical test results for their experiments. It will also be a safe environment to test underwater drone concepts, as well.

Lastly, five “Trees of Life” will be built around campus. These will consist of vertical wind turbines combined with solar panels that serve as a sustainable energy source for the campus and surrounding community. In addition to this practical use, they will serve as a symbol of this campus’ goals of creating practical and sustainable solutions to modern problems.

Table of Contents

Cover……..1

Letter of Submission……..2

Executive Summary……...3

Design #1……...5

Design #2……...7

Design #3……...9

Design #1

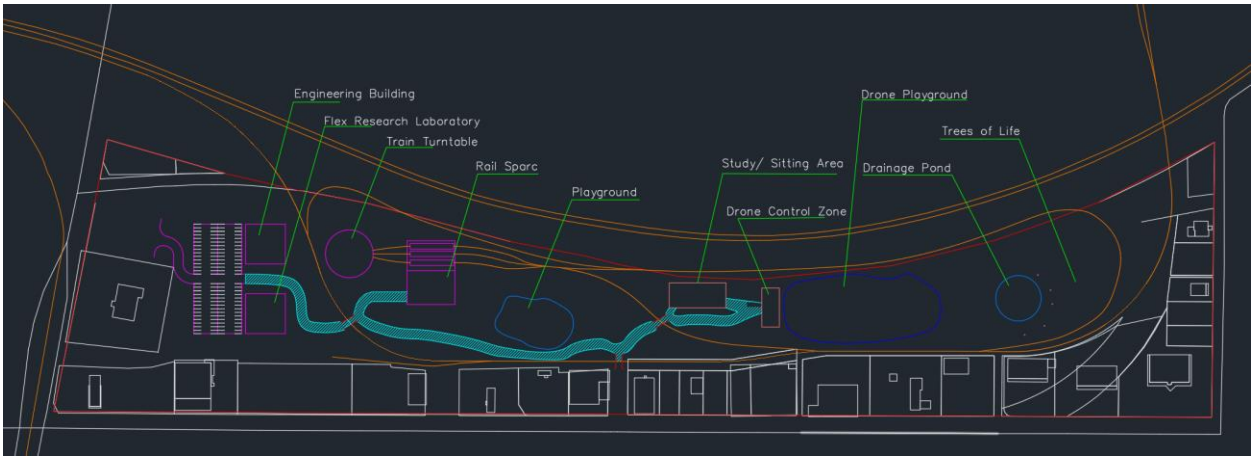

Figure 1. Design #1 Layout

Site Plan

The figure above shows the general layout of one potential design of the MARVEL Campus. The majority of the pre-existing buildings along the border roads are preserved; only several industrial facilities and warehouses in the center of the property were removed. The pre-existing railroad loop that is on-site will be utilized for campus research. As such, internal rail connections are added in the central part of the campus.

All road traffic will enter from East Main Avenue using a preexisting connection to a restaurant in the Northwest section of the property. Parking will consist of two lots situated directly next to the Engineering Education building and the Flexible Research Laboratory. A path leading to the central part of the campus will pass between the two buildings.

The central section of the campus will primarily consist of the RailSparc building, which will have three attached train workshops with direct rail connections. Additionally, there will be a large turntable that will allow for the convenient rotation of locomotives for use on the test tracks. This turntable can also serve as a scenic study area or gathering location. There is also space for a communal area such as a playground in this area, which will create some additional liveliness to the campus and increase its contribution to the

quality of life of the surrounding area. There is a southern walking connection to several local businesses such as Tony's Ice Cream, which will encourage students and staff to visit these locations due to their convenience.

The east section of the campus will include the secondary functions of the campus, including the drone playground and the water research pond. There will also be a nearby study area for students to work in an outdoor environment. This will also provide an area to sit and spectate any drone usage taking place in the designated locations. Bordering the water pond will be the five trees of life, which serve as the western border to the current campus plans.

Design #2

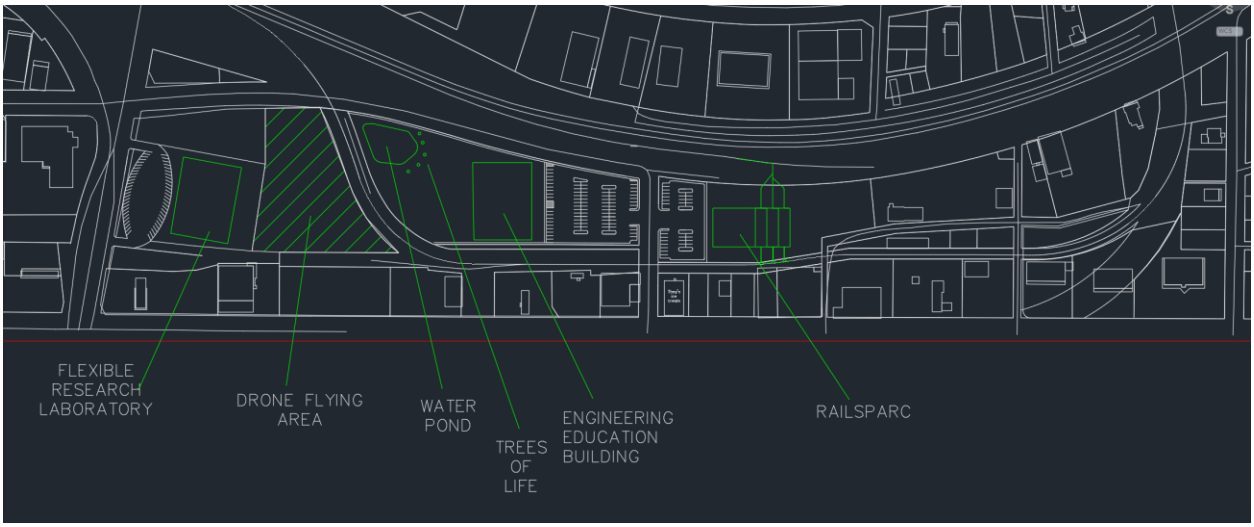

Figure 2. Design #2 Layout

Site Plan

The figure above shows the general layout for a second potential design for the MARVEL Campus. Like with the previous design, most pre-existing occupied buildings are preserved, however one restaurant on the west side is converted into a parking lot in this design that is not in the previous one. Much less additional rail track is needed in this design.

The western side of the campus will be used for the Flexible Research Laboratory and drone testing. Since it will have its own designated parking lot, students and faculty attending the campus solely for work in these areas will not need to crowd in with the rest of the campus attendees.

The central section of the campus will consist of the water pond and the Engineering Education Building. The water pond is still relatively close to the research laboratory, allowing for efficient movement between the two for testing in and out of the lab. Any relevant testing equipment can also be stored in the Engineering Education Building, which is close by as well. The largest parking lot on campus can be

found here and can serve either as general parking or designated parking for the Engineering Education Building. The Trees of Life can also be found bordering the water pond.

Across the street is another parking lot and the RailSparc building. Like in the previous design, the RailSparc building connects to the pre-existing railway on site. Unlike the previous design, the train depots themselves complete the circuit, which will mean less additional track needs to be installed on-site. The nearby parking lot can be either designated for this building or used as general parking.

Design #3

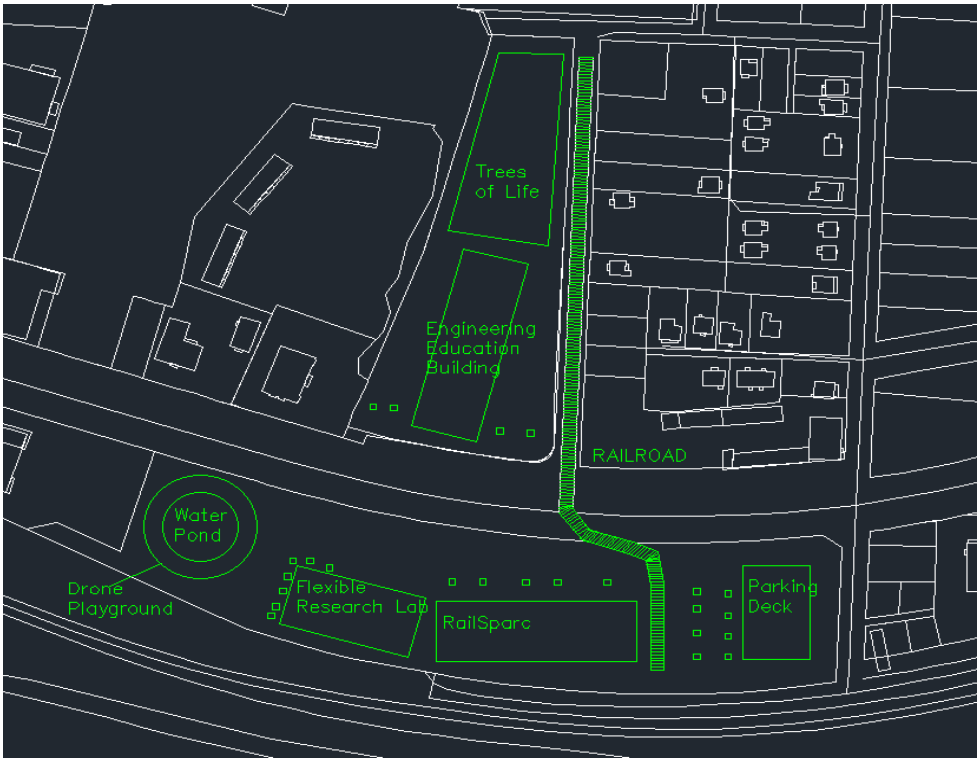

Figure 3. Design #3 Layout

Site Plan

The figure above shows the general layout for a third potential design for the MARVEL Campus. It is located just above and across the road from where the other two site plans are. Unlike the previous two designs, a couple pre-existing occupied buildings are not preserved. A dollar general, a business called Maaco, a building at 495 E Long Ave, Mr. Nobody, Gastonia Pickers Antique Mall and Auctions, and a building at 323 N Avon St will have to be removed.

The water pond is located on the western side of the campus, and the area outside of it will be used for drone testing. The northern side of the campus will be used for the Flexible Research Laboratory. There are existing parking lots in these areas that can be used for faculty and staff. Students will have to park in

the parking deck, this is not an issue for those who need to get to the engineering education building or the trees of life but could be a slight inconvenience for those who want to visit the pond and drone playground.

The central section of the campus will consist of the flexible research lab, RailSparc, and the start of the railroad. The water pond is still relatively close to the research laboratory, allowing for efficient movement between the two for testing in and out of the lab. Any relevant testing equipment can also be stored in the Engineering Education Building, which is close by as well. The largest parking lot on campus can be found on the eastern side of the campus, adjacent to the start of the railroad. The location of the parking deck and railroad route allows for easy access to every part of the campus except for the water pond and drone playground. These are located 800ft away, to the west of the parking deck.

The northern side of the campus is the engineering education building and the trees of life. The railroad also extends across the road and to the northernmost side of this part of the campus. This could be slightly inconvenient for collaborative research linking the pond, playground, and trees of life. It is convenient for an aesthetic standpoint and uses an area of the land that does not have to be demolished.